



Project #3: Robot Arm

SWE Projects - Spring 2025 x CAI

Project Lead: Tanya Koganti

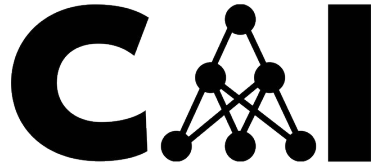




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01

Resources/ Materials

Know all your components !



Resources

- 3D printed and 4 Servos
- Mini size
- Cute and Super Simple

Color Sensing:

- TCS200 Color Sensor
- Color Sensor Code
- Robot Arm with Color Sensing



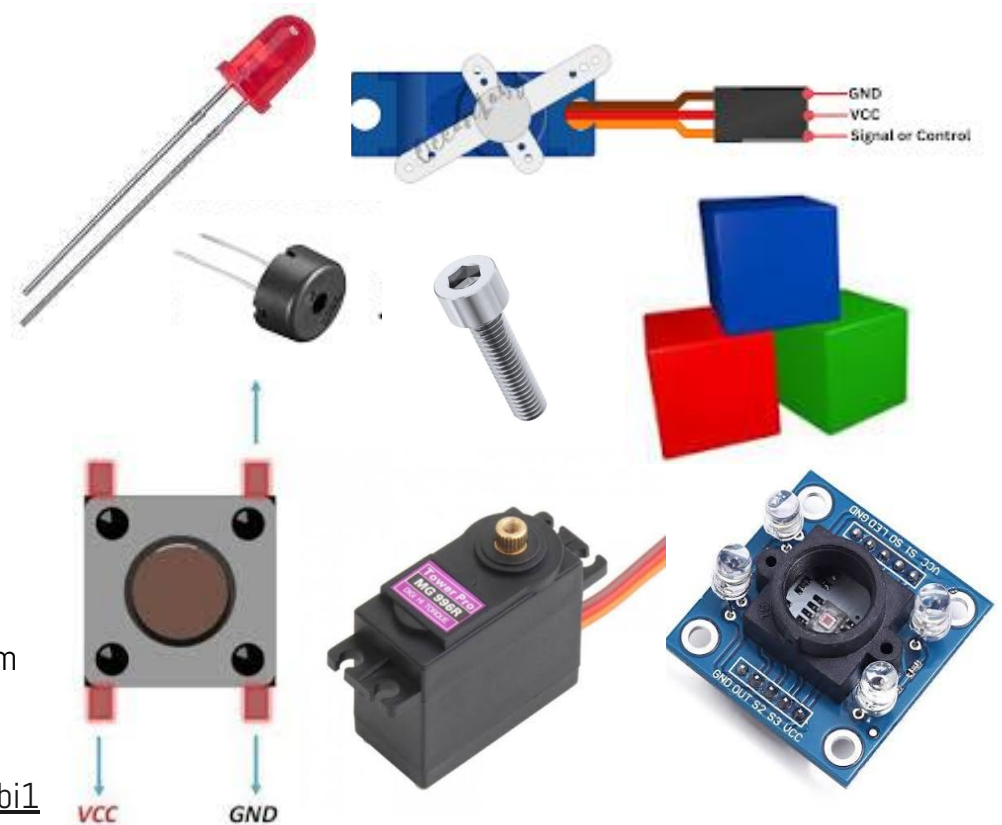
Challenge Extension

- Pick and Place Challenge
- Stack Multiple Objects
- Sort by Multiple Colors
- Increase Picking Accuracy
- Reduce Cycle Time
- Autonomous Workstation
- Multi-Object Sorting



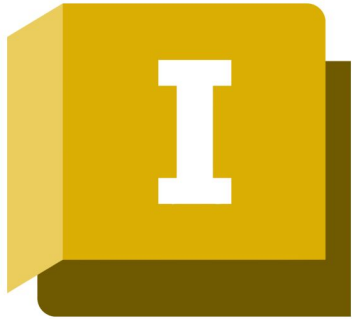
Know your Materials

- Elegoo Nano
- Breadboards & Jumper Wires
- Normal USB cable with data transfer
- Resistors (4.7 Kohms) for color sensor data clean
- Flat Wires (Different than Jumper)
- Black Electrical Tape for the line
- Servo motors e.g. MG995 (Base)
- Mini servo motors e.g. MG90 9G (Arm Links)
- TCS3200 Color Sensor
- 3d Filament Spools (Base, Body, Hand)
- Screws for Servos for servos - M2 9mm
- Screw for Attaching Base and Robot Arm - M3 14mm



Bill of Materials:

<https://docs.google.com/spreadsheets/d/119BSRgEpRg6bi15PObTLrIC4lChy02Y8APw1iigMU-U/edit?usp=sharing>



02

Design/CAD/ Mechanical

Plan out what you want to build !



Three Stages of Mechanical Prototyping



1. Design

Create sketches and plan the Robot Arm's features, layout, and interactions.



2. CAD

Develop a 3D model and integrated all electronic and mechanical components.



3. 3D Print

Fabricate and test the prototype, refining the design through multiple iterations.

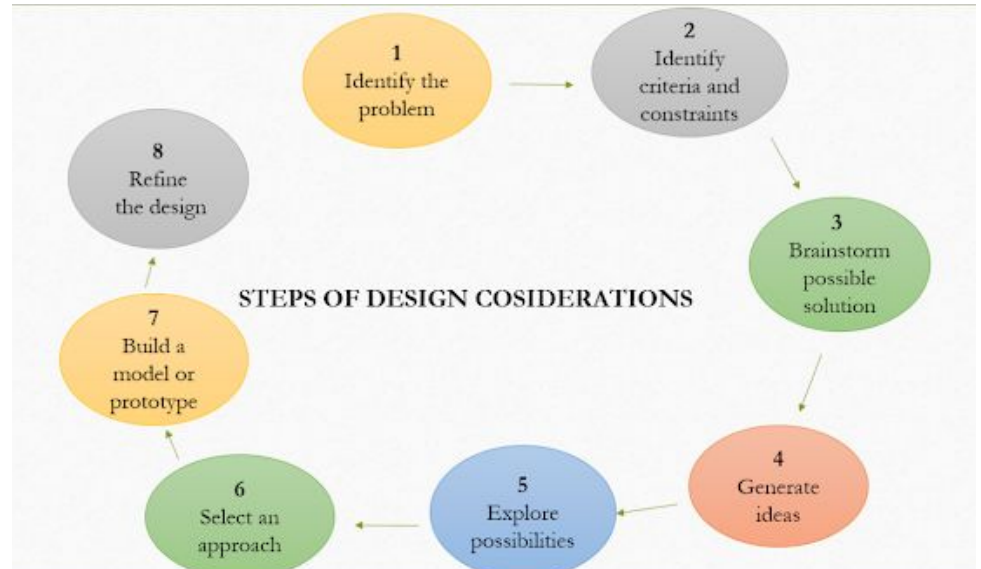
Engineering Design Constraints & Considerations

Design Considerations

Where will each component fit?
 How do we route wires cleanly?
 Will the design be easy to assemble?
 Can the structure support the hardware?
 How do we make the design compact and functional?

Mechanical Planning

Measuring motors, sensors, and batteries
 Planning mounting locations
 Designing around real component dimensions
 Balancing aesthetics and functionality



Students learned that engineering design starts long before fabrication.

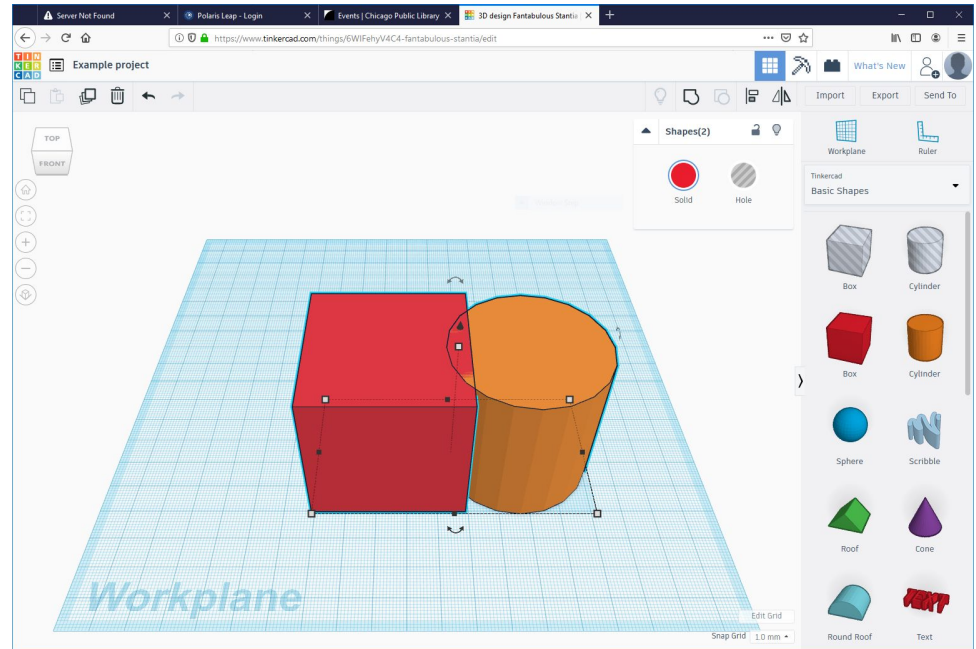
CAD - Finalize Design/3D Printing

CAD Development

- Creating initial sketches and part layouts
- Iterating designs based on component fit
- Learning tolerances and clearances
- Preparing models for fabrication

3D Printing & Prototyping Planning

- Exporting STL files for printing
- Testing printed parts for fit/function
- Adjusting designs after failed prints
- Understanding real-world manufacturing constraints
- Turning ideas into physical prototypes



Members gained hands-on experience with the full prototyping workflow.



Inventor or TinkerCAD

Inventor - AutoDesk



- Similar to SolidWorks
- Industry-standard CAD software
- Great introduction to mechanical design workflows
- Helps build technical portfolio projects
- Best for students wanting deeper CAD experience

Download:

<https://www.autodesk.com/education/edu-software/overview#INVPROSA>

TinkerCAD - AutoDesk



- Beginner-friendly CAD platform
- Easy entry point for first-time designers
- Quick learning curve for workshops/projects
- Great for Arduino + 3D printing integration
- Helps students gain confidence before advanced CAD

3D Design TinkerCAD:

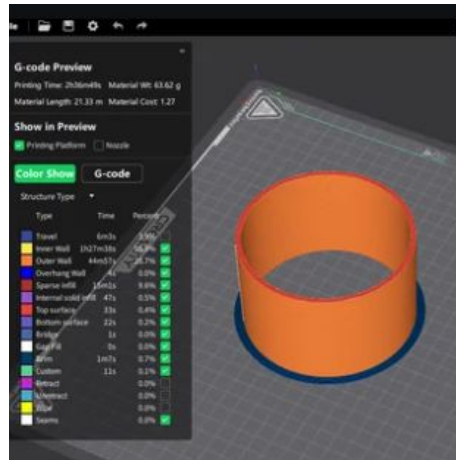
<https://www.tinkercad.com/3d-design>

These softwares chosen to make CAD more accessible for students of all experience levels.

3D Printing Slicer Software (CAD → 3D Printer)

CAD are in .STL Files
3D Printers can only read G-code

Slicer software converts .STL files to G-code instructions



Basic Slicer Workflow

- Import .STL file into slicer software
- Position model flat on the print bed
- Optimize orientation for stronger/more successful prints
- Scale or resize if necessary
- Add supports for overhangs or floating sections
- Check layer preview before printing
- Export final G-code to printer

Good Printing Practices

- Minimize unnecessary supports
- Avoid tall unstable orientations
- Keep large flat surfaces on the print bed
- Watch for unsupported overhangs
- Verify print time and material usage before printing



Ender3 Printer → Creality Slicer

Different printers may use different slicer software.

SWE projects used the Ender 3 printer with Creality Slicer.

Download Slicer Software "Creality Print":

<https://www.creality.com/download/ender-3-3d-printer>



03

Electrical/ Schematics

Wire it up, No wires = No programming...

Black Box Diagrams - Visualization

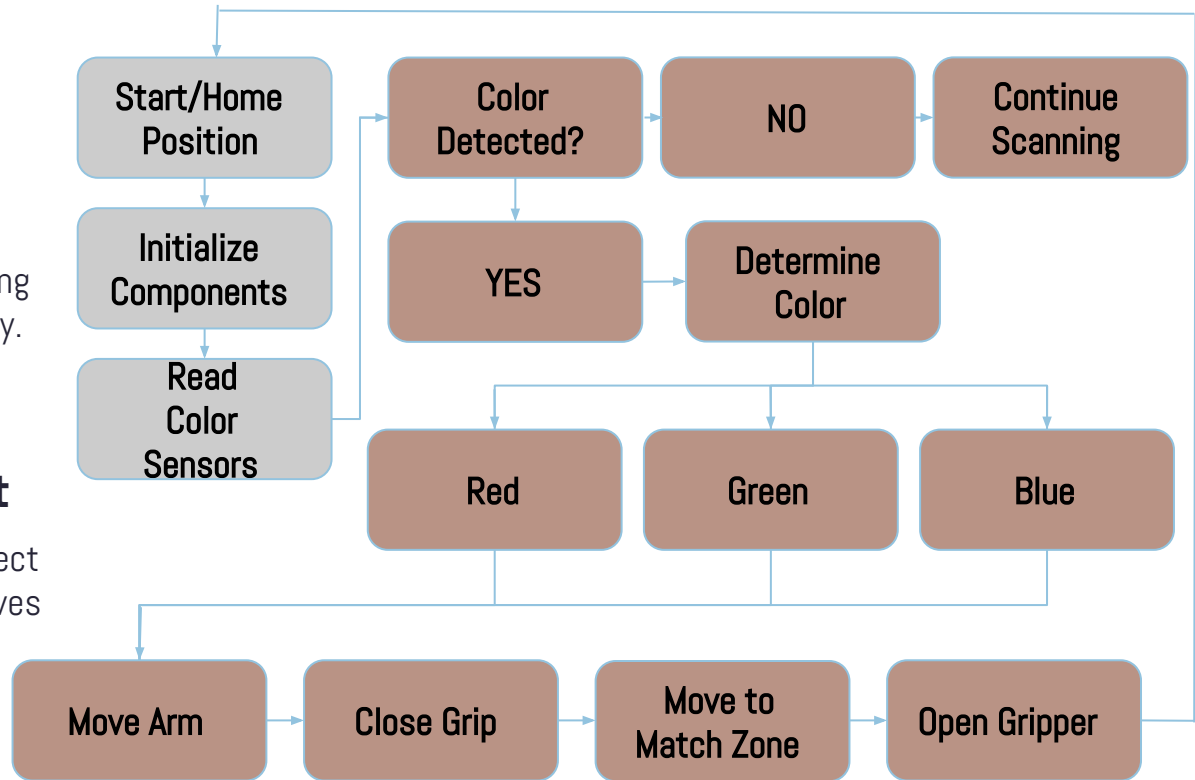
Black Box Diagram



Explains what the system does by showing inputs, outputs, and the system boundary.

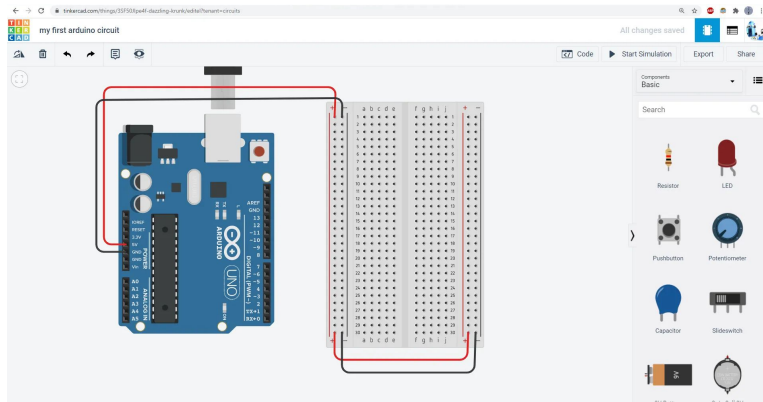
Flow Diagram/Flow Chart

Illustrates how the robot arm detects object color, chooses the correct action, and moves the servos to sort or place the object.



Breadboard

TinkerCAD



Resources

Reading:

<https://www.build-electronic-circuits.com/breadboard/>

Videos:

https://youtu.be/xs8No_rg40U?si=nRS4dQbBDzABbNn

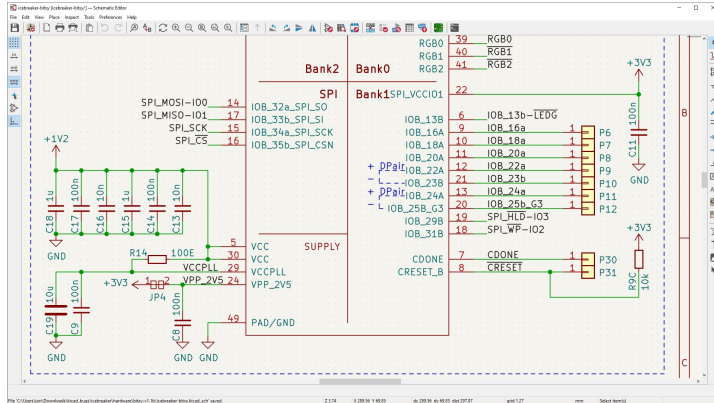
Intro Slides: https://docs.google.com/presentation/d/1yD5nXMJWPMtZPHhtXyCd_YX-FAF2YCUzgcZnmeqP8_c/edit?usp=sharing



KiCAD or TinkerCAD - Schematic

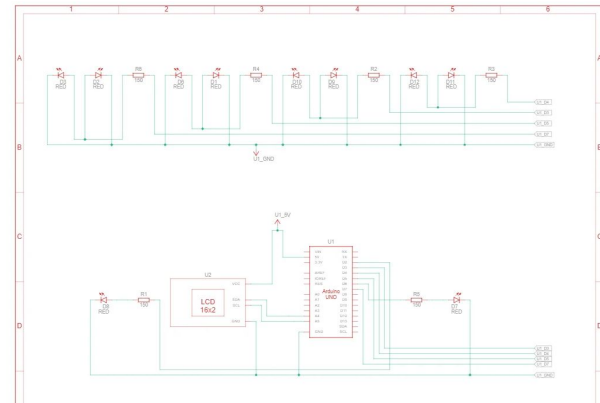
KiCAD

Allows for PCB development
Good industry skill to learn



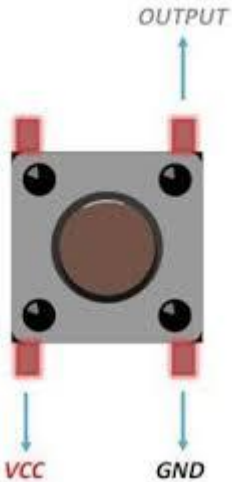
TinkerCAD

Super Beginner friendly
Easy to learn



Components (Input)

Push Button



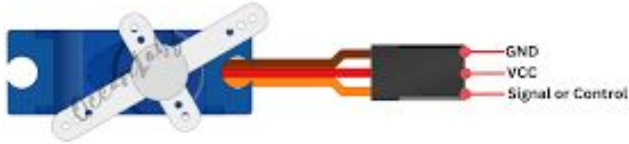
Provides user input to start, stop, or reset the robot arm. It can also be used to initiate the color-sorting routine or trigger a predefined movement sequence.

Servo Motor (MG995)



Provides precise angular movement for each joint of the robot arm. The Arduino controls the servo motors to accurately position the base, arm, and gripper for picking up and sorting objects.

Components



Servo Motor

Provides precise angular movement for each joint of the robot arm. Multiple servo motors work together to position the arm, rotate the joints, and control the gripper for accurate object manipulation.



Color Sensor (TCS3200)

Detects the color of nearby objects by measuring the intensity of reflected red, green, and blue light. The Arduino uses this information to identify objects and sort to appropriate spot.

Components (Output)

Passive Buzzer



Buzzer (Passive)

Produces customizable tones and melodies using different frequencies. It allows the Robot arm to play sounds for completing certain actions.

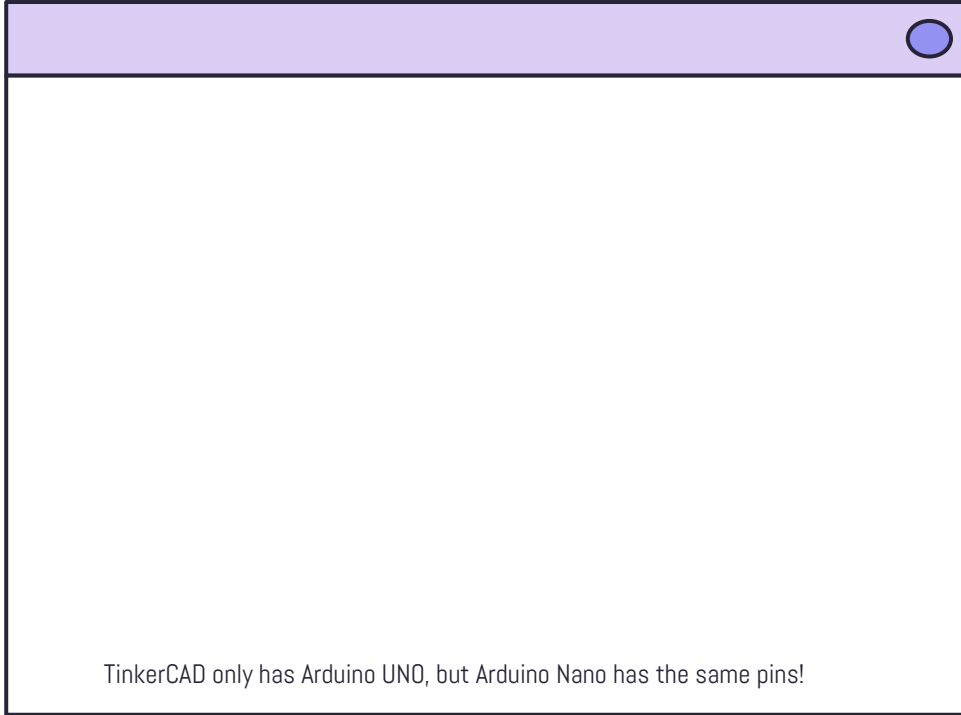
Active Buzzer



Buzzer (Active)

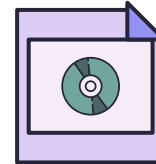
Generates a simple alert sound when powered. It is used for quick notifications, alarms, and status indicators.

Sample Circuit - Customize your Own !!



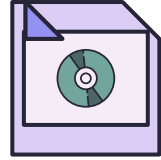
Nano

main microcontroller,
reading sensor inputs
and controlling outputs



Jumper Wires

Connect the Arduino,
breadboard, and
electronic components



Power

Distributes 5V and
GND from the Arduino
to the breadboard,



Resistors

Limit current to
protect components
and ensure safety



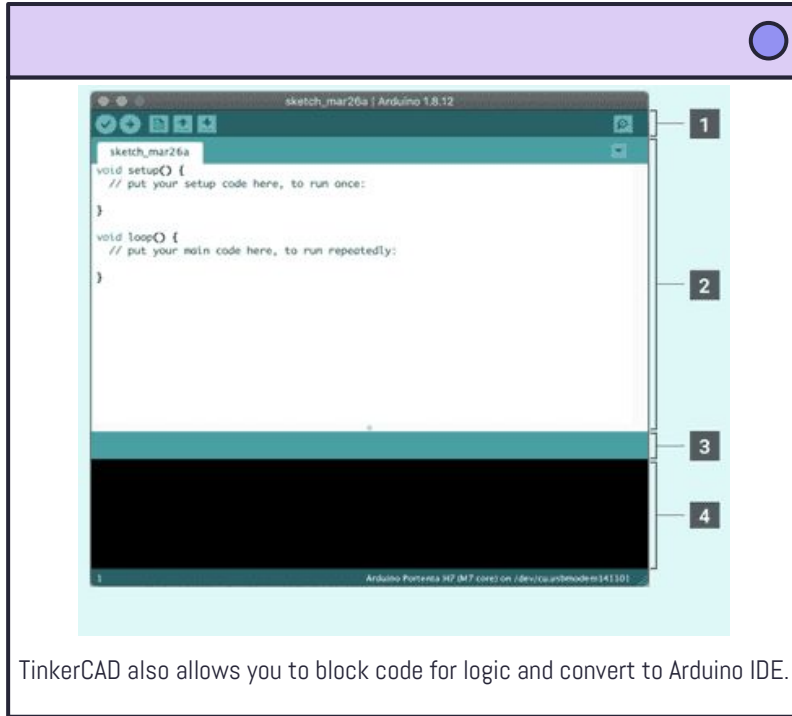
04

Programming

Watch your build come to life !!



Arduino IDE (C++) -

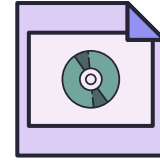


TinkerCAD also allows you to block code for logic and convert to Arduino IDE.



Download

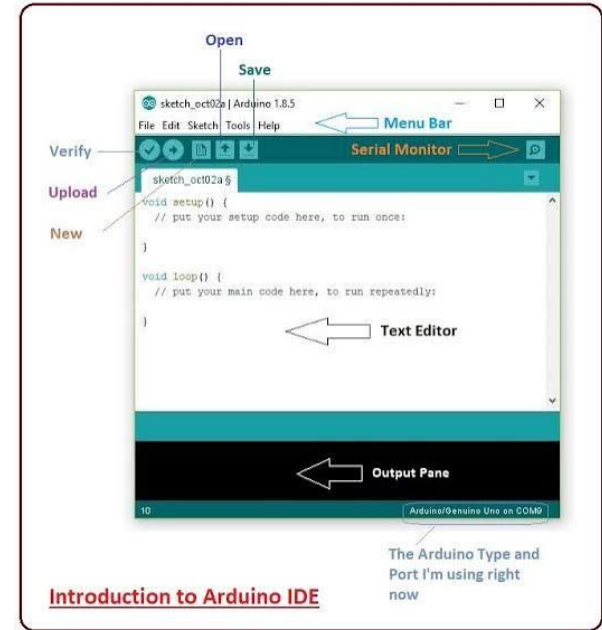
<https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE>



Arduino IDE Function Library

<https://docs.arduino.cc/language-reference/>

Arduino IDE - Must Know



Arduino IDE Intro: <https://docs.arduino.cc/software/ide-v2/tutorials/getting-started-ide-v2/>



Arduino IDE - Parts

The screenshot shows the Arduino IDE window titled "Blink | Arduino 1.0". The code editor contains the following text:

```
/*
 * Blink
 * Turns on an LED on for one second, then off for one second, repeatedly.
 *
 * This example code is in the public domain.
 */

void setup() {
  // initialize the digital pin as an output.
  // Pin 13 has an LED connected on most Arduino boards:
  pinMode(13, OUTPUT);
}

void loop() {
  digitalWrite(13, HIGH); // set the LED on
  delay(1000);            // wait for a second
  digitalWrite(13, LOW);  // set the LED off
  delay(1000);            // wait for a second
}
```

At the bottom of the window, a status bar indicates "1" and "Arduino Uno on /dev/tty.usbmodem621".

void setup ()

Setup Function

Runs once when the Arduino powers on. Used to initialize components such as sensors, servos, buzzers, and the LCD display before the Desk Buddy begins operating.

void loop ()

Main Program Loop

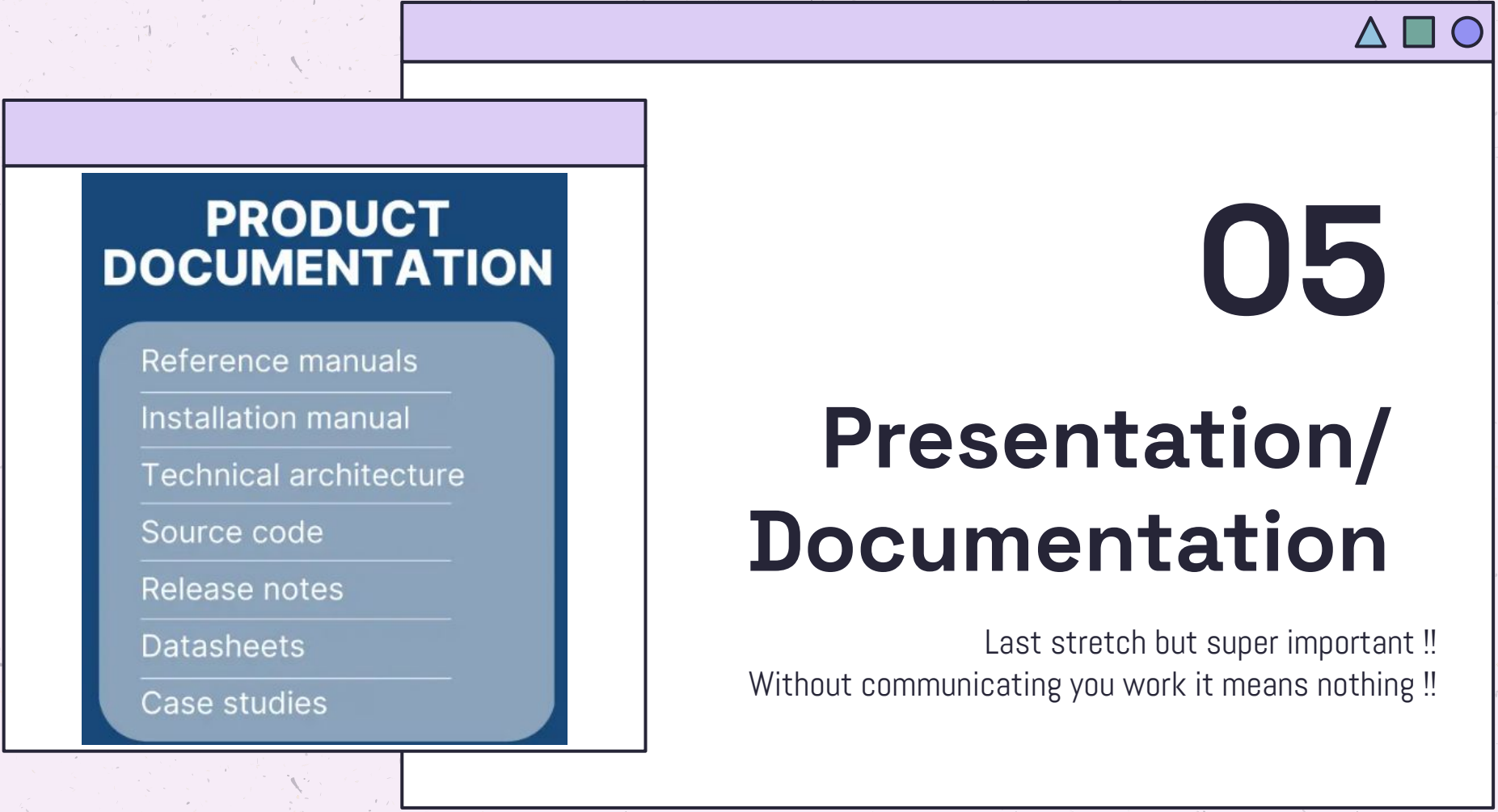
Runs continuously after `setup()`. It repeatedly checks sensor inputs and updates the Desk Buddy's actions, displays, sounds, and movements.

Variables

Store information used throughout the program, such as temperature readings, motion detection status, servo positions, and user settings.

Functions

Reusable blocks of code that perform specific tasks. Functions help organize the program and simplify actions like displaying messages, playing sounds, or moving servos.



PRODUCT DOCUMENTATION

Reference manuals

Installation manual

Technical architecture

Source code

Release notes

Datasheets

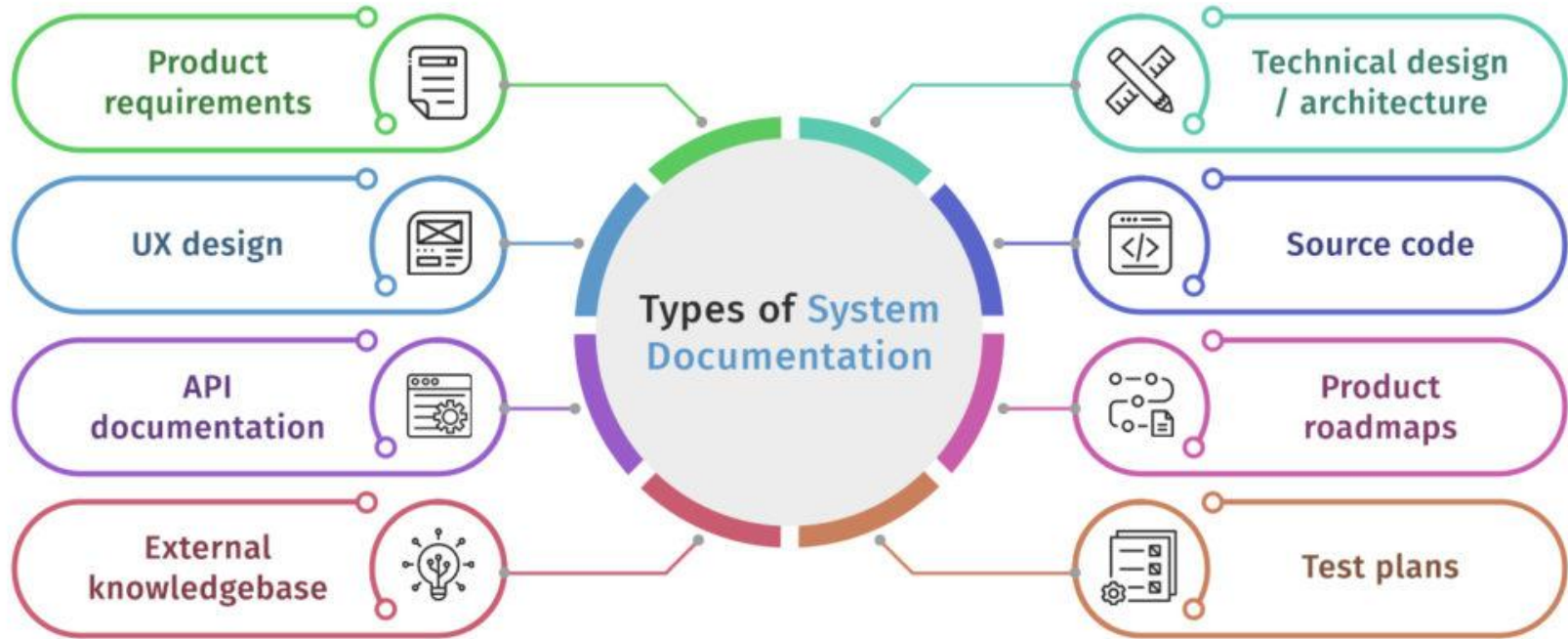
Case studies

05

Presentation/ Documentation

Last stretch but super important !!
Without communicating your work it means nothing !!

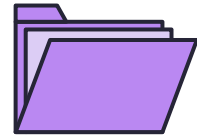
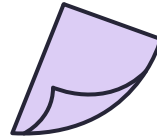
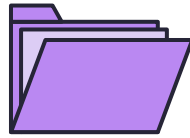
Product Based Documentation !!



<https://technicalwriterhq.com/documentation/product-documentation/>



Presentation - 4 parts



Design/CAD

CAD Files

Schematics

Circuit Schematic
File

Program w/ Comments

Arduino IDE Code
w/ Comments

Pictures of final Product

Pictures/Videos

Presentation

Presentation Template:

SWE Project Showcase (Demo) - Awards !!



Please have your
Project + Slides ready for our
Inaugural SWE Project Showcase !!

Thursday, April 24th
6:00 - 8:00pm
Arch 310

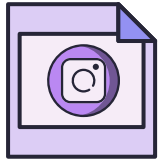


Thanks!

Do you have any questions?



Please Contact Me !!
Project.swe.uh@gmail.com
@Tako_124 on Discord



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semester:

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Pt 1. Troubleshooting Help - Common



Servo not moving

Verify power connection

Check signal pin

Ensure servo is not stalled



Arm misses or drops objects

Calibrate servo angles

Adjust gripper position

Reduce movement speed



Color sensor not detecting objects

Adjust lighting conditions

Calibrate RGB thresholds

Verify sensor wiring



Pt 2. Troubleshooting Help - Common



Wrong color detected

Calibrate RGB values

Reduce ambient light

Position object closer to sensor



Servos jitter

Use external 5V supply

Secure loose connections

Reduce servo load



Robot doesn't sort correctly

Verify color thresholds

Check servo positions

Test pick-and-place sequence

Further Learning

- D

Keep Learning !!

Future Improvements

- Add More Degrees of Freedom (DOF)
- Automatic Color Sorting
- Bluetooth/Wi-Fi Control
- Object Detection with Camera
- OLED/LCD Status Display
- Voice or Gesture Control
- Inverse Kinematics
- AI Vision Integration

